VANISHING OF CECH H1 ON A COFINAL SYSTEM OF COVERINGS MAKES THE MAP ON SECTIONS SURJECTIVE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, ending with the author's own closing phrase, which is retained verbatim.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space with O_X-modules. The conclusion is the same and the ringed-space statement is the special case of this arbitrary-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Cech cohomology H^1 of a covering with coefficients in a sheaf. Same chapter, the definition of the Cech complex of a covering U of an object U and of its first cohomology H^1(U, F). Supplies the vanishing hypothesis and the cocycle statement applied to the sections s_{i_0 i_1} in the counted proof.
- Cofinal system of coverings of an object of a site. Same chapter, the definition of a cofinal system of coverings. Allows the covering satisfying the lifting condition (b) to be refined by one for which the vanishing (a) also holds.
- Sheaf condition for a covering and gluing of sections. Same chapter, the sheaf axiom for a covering in a site. Supplies the final gluing step, in which the corrected sections s_i - t_i glue to a section of G over U.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let*

$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$

be a short exact sequence of abelian sheaves on $\mathcal{C}$. Let U be an object of $\mathcal{C}$. If there exists a cofinal system of coverings $\mathcal{U}$ of U such that $\check{H}^1(\mathcal{U}, \mathcal{F}) = 0$, then the map $\mathcal{G}(U) \to \mathcal{H}(U)$ is surjective.

Proof. Take an element $s \in \mathcal{H}(U)$. Choose a covering $\mathcal{U} = \{U_i \to U\}_{i \in I}$ such that (a) $\check{H}^1(\mathcal{U}, \mathcal{F}) = 0$ and (b) $s|_{U_i}$ is the image of a section $s_i \in \mathcal{G}(U_i)$. Since we can certainly find a covering such that (b) holds it follows from the assumptions of the lemma that we can find a covering such that (a) and (b) both hold. Consider the sections

$$s_{i_0 i_1} = s_{i_1}|_{U_{i_0} \times_U U_{i_1}} - s_{i_0}|_{U_{i_0} \times_U U_{i_1}}.$$

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Since s_i lifts s we see that $s_{i_0 i_1} \in \mathcal{F}(U_{i_0} \times_U U_{i_1})$. By the vanishing of $\check{H}^1(\mathcal{U}, \mathcal{F})$ we can find sections $t_i \in \mathcal{F}(U_i)$ such that

$$s_{i_0 i_1} = t_{i_1}|_{U_{i_0} \times_U U_{i_1}} - t_{i_0}|_{U_{i_0} \times_U U_{i_1}}.$$

Then clearly the sections $s_i - t_i$ satisfy the sheaf condition and glue to a section of $\mathcal{G}$ over U which maps to s. Hence we win. $\square$